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§ 1. Whole functions.

The next object of consideration is whole rational functions, or, briefly, whole functions of one variable. By these we understand expressions of the following form:

$$f(x) = a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \cdots + a_{n-1}x + a_n, \quad (1)$$

where n is an integral exponent, the degree of the function $f(x)$. The degree is a natural number. Sometimes, however, it is also useful to speak of whole rational functions of degree 0, by which a constant is understood. x is called the variable, $a_0, a_1, a_2, \dots, a_{n-1}, a_n$ the coefficients. Both x and $a_0, a_1, \dots, a_n$ are symbols for indeterminate quantities, with which one proceeds according to the rules of literal calculation, and for which definite numerical values may also be substituted under some circumstances (cf. the Introduction).

If the function $f(x)$ is written in the way shown in (1), we call it ordered according to descending powers of x . The summands may be arranged in any other order; for example, the function may also be ordered according to ascending powers of x :

$$f(x) = a_n + a_{n-1}x + \cdots + a_1x^{n-1} + a_0x^n.$$

By addition, subtraction, and multiplication of whole rational functions there again arise whole rational functions. The rules of literal calculation directly give the laws of formation of these new functions.

In addition, the coefficient of any power x^ν in the sum is equal to the sum of the coefficients of x^ν in the individual summands. The degree of the function that arises is equal to the highest of the degrees of the summands, and can decrease only in the special case in which the highest degree occurs in several summands and the sum of the coefficients of the highest powers is zero.

In the multiplication of two or more whole rational functions there arises a function whose degree is equal to the sum of the degrees of the factors.

To survey the law of formation of the coefficients for the product, put

$$\begin{aligned} A(x) &= a_0x^m + a_1x^{m-1} + a_2x^{m-2} + \cdots, \\ B(x) &= b_0x^n + b_1x^{n-1} + b_2x^{n-2} + \cdots, \end{aligned} \quad (2)$$

$$A(x)B(x) = C(x) = c_0x^{m+n} + c_1x^{m+n-1} + c_2x^{m+n-2} + \cdots \quad (3)$$

and obtain

$$\begin{aligned} c_0 &= a_0b_0, \\ c_1 &= a_0b_1 + a_1b_0, \\ c_2 &= a_0b_2 + a_1b_1 + a_2b_0, \\ &\cdots, \end{aligned}$$

or, in general notation, if ν denotes one of the numbers $0, 1, 2, \dots, m+n$,

$$c_\nu = a_0b_\nu + a_1b_{\nu-1} + a_2b_{\nu-2} + \cdots + a_{\nu-1}b_1 + a_\nu b_0, \quad (4)$$

where all a whose index is greater than m , and all b whose index is greater than n , are to be set equal to zero. For c_ν is the coefficient of $x^{m+n-\nu}$, and hence $c_\nu x^{m+n-\nu}$ arises by multiplying all

terms of the form $a_\mu x^{m-\mu}$ by all terms of the form $b_{\nu-\mu} x^{n-\nu+\mu}$ and then adding. If a_0 and $b_0 = 1$, then also $c_0 = 1$.

If in all factors of such a product the highest power of x has coefficient 1, then in the product too the coefficient of the highest power is 1.

From these rules for calculating with whole functions it follows that the rules of calculation expressed in formulae such as $ab = ba$, $(ab)c = a(bc)$, $(a+b)c = ac + bc$ and the like also hold when a, b, c are whole functions of x , and in the sense that whole functions are to be regarded as equal only when equal powers have equal coefficients.

By whole functions of several variables we understand sums of products of whole, positive powers of the variables $x, y, z \dots$ with arbitrary coefficients that count as constant. One can think of these functions as arising from functions of one variable x when the coefficients $a_0, a_1 \dots a_n$ occurring in them are themselves regarded as whole rational functions of other variables $y, z \dots$. Thus functions of m variables arise from functions of $m-1$ variables. All terms of such a function are of the form $x^r y^s z^t \dots$, multiplied by a coefficient; and if the function is properly collected, then each combination of the exponents $r, s, t \dots$ occurs only once. Two whole functions so ordered are regarded as equal only when they contain the same products $x^r y^s z^t \dots$ with the same coefficients.

§ 2. A theorem of Gauss.

We shall at once make an application of the multiplication rule for two whole rational functions, in order to prove a theorem of Gauss which will be useful to us later, but which here is to serve as an introduction to the mode of calculation and as an example.¹

We consider here the case in which the coefficients in the functions $A(x), B(x)$ are whole numbers, so that, by (4), the coefficients of $C(x) = A(x)B(x)$ are also whole numbers.

If all integral coefficients $a_0, a_1, \dots, a_m$ of a function $A(x)$ have no common divisor, then the function $A(x)$ is called an original or primitive function, and the theorem we wish to prove is:

If $A(x)$ and $B(x)$ are primitive functions, then their product $C(x)$ is also a primitive function.

The proof follows almost immediately from looking at formula (4).

For if all coefficients $c_0, c_1, c_2 \dots c_{m+n}$, as given in (4), had a common divisor greater than 1, then there would also have to be at least one prime number which divides all these coefficients. Let p be such a prime number. By the assumption that $A(x), B(x)$ are primitive, this prime can divide neither all $a_0, a_1 \dots a_m$ nor all $b_0, b_1 \dots b_n$.

Suppose now that p divides

$$a_0, a_1 \dots a_{r-1}, \quad \text{but not } a_r,$$

and divides

$$b_0, b_1 \dots b_{s-1}, \quad \text{but not } b_s.$$

Then, by assumption, $r \leq m$, and r may also be zero if p already does not divide a_0 . Likewise $s \leq n$.

If now, by (4), we form c_{r+s} and arrange it in the following way,

$$\begin{aligned} c_{r+s} = & a_r b_s + a_{r-1} b_{s+1} + a_{r-2} b_{s+2} + \dots \\ & + a_{r+1} b_{s-1} + a_{r+2} b_{s-2} + \dots, \end{aligned} \tag{1}$$

one sees at once that c_{r+s} cannot be divisible by p , as had been assumed; for the first term $a_r b_s$ is not divisible by p , while all other terms, since they are multiplied by one of the coefficients $a_0, a_1 \dots a_{r-1}, b_0, b_1 \dots b_{s-1}$, are divisible by p . Thus the assumption that $C(x)$ is not primitive while $A(x)$ and $B(x)$ are primitive leads to a contradiction.

¹Gauss, *Disquisitiones arithmeticae*, Art. 42.

This theorem can be transferred to functions of several variables. We call a whole rational function of m variables with integral coefficients original or primitive if the coefficients have no common divisor, and we state the theorem that the product of two primitive functions is again a primitive function.

To see its truth, we need only take, in the argument carried out above, the coefficients $a_0, a_1, a_2, \dots, b_0, b_1, b_2, \dots$ not as whole numbers, but as whole rational functions of $m-1$ variables $y, z, \dots$, and call such a function divisible by a prime number p when all its coefficients are divisible by p . If we then suppose that the theorem to be proved has already been proved for functions of $m-1$ variables, formula (1) gives its correctness for functions of m variables, hence its general validity by complete induction, or by the conclusion from $m-1$ to m .

An imprimitive function is one whose integral coefficients all have a common divisor. We call this divisor the divisor of the function. Then we can also express the theorem just proved as follows:

The divisor of a product of two whole functions is equal to the product of the divisors of the two functions.

For if PA and QB are two whole functions with divisors P and Q , then A and B are primitive functions. Hence $AB = C$ is also a primitive function, and PQ is the divisor of the function $PA \cdot QB = PQC$.

Insofar as it refers to functions of one variable, we can give the theorem the following form, without changing its content essentially; in this form it is especially useful. Let

$$\varphi(x) = x^m + \alpha_1 x^{m-1} + \alpha_2 x^{m-2} + \dots + \alpha_m,$$

$$\psi(x) = x^n + \beta_1 x^{n-1} + \beta_2 x^{n-2} + \dots + \beta_n$$

be two whole rational functions in which the highest powers of x have coefficient 1, while the remaining coefficients are rational numbers. Then, in the product

$$\varphi(x)\psi(x) = x^{m+n} + \gamma_1 x^{m+n-1} + \gamma_2 x^{m+n-2} + \dots + \gamma_{m+n},$$

the coefficients γ cannot all be whole numbers if the coefficients α, β in $\varphi(x)$ and $\psi(x)$ are not all whole numbers.

For let the smallest common denominator of the coefficients α of φ be denoted by a_0 , and that of the coefficients β of ψ by b_0 . Then $a_0\varphi(x) = A(x)$ and $b_0\psi(x) = B(x)$ are primitive functions of x . If the coefficients γ were whole numbers, the product $a_0b_0\varphi(x)\psi(x)$ would have divisor a_0b_0 and, if a_0 and b_0 were not both equal to 1, would therefore not be primitive; but this would contradict the theorem proved above.

§ 3. Division.

Let, as before,

$$\begin{aligned} A &= A(x) = a_0 x^m + a_1 x^{m-1} + \dots, \\ B &= B(x) = b_0 x^n + b_1 x^{n-1} + \dots \end{aligned} \tag{1}$$

be two whole rational functions of x ; but now suppose that $m \geq n$ and that a_0 and b_0 are different from zero. Then the difference

$$A - \frac{a_0}{b_0} x^{m-n} B \tag{2}$$

is also a whole rational function of x , but its degree is smaller than m , since the highest power in the two terms of the difference has the same coefficient and therefore cancels. We put this difference equal to

$$A' = A'(x) = a'_0 x^{m'} + a'_1 x^{m'-1} + \dots, \quad m' < m. \tag{3}$$

If now m' is still $\geq n$, then in (2) we can put A' in place of A and repeat the same reasoning.

This gives a chain of equations:

$$\begin{aligned} A - \frac{a_0}{b_0}x^{m-n}B &= A', \\ A' - \frac{a'_0}{b_0}x^{m'-n}B &= A'', \\ A'' - \frac{a''_0}{b_0}x^{m''-n}B &= A''', \\ &\dots, \end{aligned} \tag{4}$$

and this chain can be continued until the degree of the function that has arisen is smaller than n . Since in the sequence of functions $A, A', A'' \dots$ the degree of each following function is lowered by at least one unit, the chain of equations (4) consists of at most $m - n + 1$ members; but it may also contain fewer members when several powers cancel simultaneously. If we add all equations (4), denote the last of the functions $A, A' \dots$ by C , and, for abbreviation, set

$$Q = \frac{a_0}{b_0}x^{m-n} + \frac{a'_0}{b_0}x^{m'-n} + \dots, \tag{5}$$

so that Q also is a whole rational function of x , then there follows

$$A = QB + C. \tag{6}$$

The operation just described, by which the functions Q, C are found from A, B , is called division. A is the dividend, B the divisor, C the remainder, and Q the quotient. The degree of the remainder is always lower than the degree of the divisor.

The coefficients of the functions Q and C are composed from the coefficients a and b by addition, subtraction, multiplication, and division. In the denominator, however, only powers of b_0 occur; and hence if $b_0 = 1$, the coefficients of Q and C are whole functions of the a and b . The highest power of b_0 which can occur in the denominator is the $(m - n + 1)$ st, since in the chain (4) the factor b_0 is added once in the denominator in each following equation. In special cases, however, the highest power of b_0 in all denominators may be a lower one.

Take, for example, for A a function of the third degree,

$$f(x) = a_0x^3 + a_1x^2 + a_2x + a_3, \tag{7}$$

and for B the so-called first derivative of $f(x)$, which is of the second degree,

$$f'(x) = 3a_0x^2 + 2a_1x + a_2. \tag{8}$$

Then one obtains

$$Q = \frac{1}{3}x + \frac{a_1}{9a_0}, \tag{9}$$

$$C = \frac{6a_0a_2 - 2a_1^2}{9a_0}x + \frac{9a_0a_3 - a_1a_2}{9a_0}. \tag{10}$$

How the calculation prescribed in the equations (4) is expediently arranged may here be assumed as known from the elements. We draw attention to the analogy between this calculation and division in the decimal number system. A function $f(x)$ represents a number written decimally if the coefficients $a_0, a_1 \dots$ are whole numbers between zero, inclusive, and 10, exclusive, and $x = 10$ is set. If one also permits numbers greater than 10 among the coefficients, then one can represent a number by $f(x)$ in various ways. This is used in the division procedure in order to avoid fractional and negative coefficients in the results in the simplest way.

§ 4. Division by a linear function.

We shall apply the general rule for division to one special case. Let the dividend be

$$f(x) = a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \cdots + a_n, \quad (1)$$

arbitrary, while the divisor is of the first degree, or, as one also says, linear. We shall also suppose the coefficient of the first power of x in the divisor to be 1, and therefore take the divisor in the simple form $(x - \alpha)$, where α remains arbitrary. In this case the quotient Q is of degree $n - 1$, and the remainder C of degree 0, that is, independent of x . Thus set

$$f(x) = (x - \alpha)Q + C. \quad (2)$$

Then C no longer contains x , and if we set

$$Q = q_0x^{n-1} + q_1x^{n-2} + \cdots + q_{n-2}x + q_{n-1}, \quad (3)$$

there follows from (2)

$$\begin{aligned} f(x) &= q_0x^n + q_1x^{n-1} + \cdots + q_{n-2}x^2 + q_{n-1}x + C \\ &\quad - \alpha q_0x^{n-1} - \cdots - \alpha q_{n-3}x^2 - \alpha q_{n-2}x - \alpha q_{n-1}, \end{aligned} \quad (4)$$

and by comparison with (1):

$$\begin{aligned} q_0 &= a_0, \\ q_1 - \alpha q_0 &= a_1, \\ q_2 - \alpha q_1 &= a_2, \\ &\cdots, \\ q_{n-1} - \alpha q_{n-2} &= a_{n-1}, \\ C - \alpha q_{n-1} &= a_n. \end{aligned} \quad (5)$$

From this one obtains

$$\begin{aligned} q_0 &= a_0, \\ q_1 &= a_0\alpha + a_1, \\ q_2 &= a_0\alpha^2 + a_1\alpha + a_2, \\ &\cdots, \\ q_{n-1} &= a_0\alpha^{n-1} + a_1\alpha^{n-2} + \cdots + a_{n-1}, \\ C &= a_0\alpha^n + a_1\alpha^{n-1} + \cdots + a_{n-1}\alpha + a_n = f(\alpha). \end{aligned} \quad (6)$$

C arises from $f(x)$ by setting $x = \alpha$, and therefore can also be denoted by $f(\alpha)$.

Accordingly we also have the formula

$$\frac{f(x) - f(\alpha)}{x - \alpha} = Q(x), \quad (7)$$

where $Q(x)$ is a whole function of degree $n - 1$.

If in the expressions (6) we put the sign x in place of the indeterminate quantity α , there arises from this a sequence of whole rational functions of x , which, if for simplicity we put $a_0 = 1$, we write as follows:

$$\begin{aligned} f_0 &= 1, \\ f_1 &= x + a_1, \\ f_2 &= x^2 + a_1x + a_2, \\ &\cdots, \\ f_{n-1} &= x^{n-1} + a_1x^{n-2} + a_2x^{n-3} + \cdots + a_{n-1}. \end{aligned} \quad (8)$$

These functions $f_0, f_1 \dots f_{n-1}$ will later render us good service. For the present we add the following remarks.

According to (8), one can express the powers $1, x, x^2 \dots x^{n-1}$ of x linearly in terms of the functions $f_0, f_1 \dots f_{n-1}$, and indeed in such a way that only integral rational combinations of the a occur in the coefficients, for instance

$$\begin{aligned} 1 &= f_0, \\ x &= f_1 - a_1 f_0, \\ x^2 &= f_2 - a_1 f_1 + (a_1^2 - a_2) f_0, \\ &\dots \end{aligned}$$

From this it follows that every whole rational function of x whose degree is not greater than $n - 1$ can likewise be expressed linearly by $f_0, f_1, \dots, f_{n-1}$ in the form

$$y_0 f_0 + y_1 f_1 + \dots + y_{n-1} f_{n-1}, \quad (9)$$

where the coefficients $y_0, y_1 \dots y_{n-1}$ are independent of x .

Thus, if $F(x)$ is any whole rational function of x , then by § 3, regarding $f(x)$ as divisor, one can set

$$F(x) = Qf(x) + y_0 f_0 + y_1 f_1 + \dots + y_{n-1} f_{n-1}, \quad (10)$$

where Q is also a whole rational function of x .

For the recurrent calculation of the functions $f_r(x)$, relation (8) gives

$$f_r(x) - x f_{r-1}(x) = a_r. \quad (11)$$